EDUEXPOSE

STUDENT ACADEMIC PERFORMANCE PREDICTOR

Machine Learning | Regression Analysis | 4,000 Student Records

PRESENTED BY

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Internship Project — Eduexpose

4,000	**91.6%**	**3.99**	**4**
Records	R^2 Score	MAE	Models

PROJECT OVERVIEW

OBJECTIVE

Build an ML regression model to accurately predict student exam scores (17-100) from study behaviour and demographic data.

DATASET

4,000 student records, 6 input features: gender, study hours, tutoring, region, attendance, parent education.

APPROACH

EDA -> Preprocessing Pipeline -> 4 Models trained -> GridSearchCV hyperparameter tuning -> SHAP explainability.

OUTCOME

Best model achieves R2 = 0.9156, MAE = 3.99, RMSE = 5.03. Strong predictive accuracy on unseen data.

DATASET & EXPLORATORY DATA ANALYSIS

Feature	Type	Range / Values
HoursStudied/Week	Numerical	0 - 16 hrs
Attendance(%)	Numerical	50 - 100%
Gender	Categorical	Male / Female
Tutoring	Categorical	Yes / No
Region	Categorical	Urban / Rural
Parent Education	Categorical	Primary / Secondary / Tertiary
Exam_Score (TARGET)	Numerical	**17 - 100**

KEY EDA FINDINGS

Study hours vs score: r = 0.82

Attendance vs score: r = 0.24

422 missing values in Parent Education

Exam scores peak in the 60-80 range

Gender split: 51% Female / 49% Male

70% students have NO tutoring

Urban outnumber Rural 60% / 40%

ML PIPELINE & METHODOLOGY

01 Data Cleaning

Remove duplicates
Impute Parent Education
422 nulls handled

02 Preprocessing

StandardScaler (numeric)
OneHotEncoder (categorical)
sklearn Pipeline

03 Model Training

Linear Regression
Decision Tree
Random Forest
Gradient Boosting

04 Tuning

GridSearchCV 5-fold CV
RF: max_depth=10
n_estimators=200

05 Evaluation

R2, MAE, RMSE
Learning curve analysis
Residual diagnostics

06 Explainability

SHAP LinearExplainer
Feature importance
Beeswarm plot

MODEL COMPARISON & RESULTS

Model	Train R2	Test R2	MAE	RMSE	CV R2 (5-fold)
* Linear Regression	0.9061	0.9156	3.99	5.03	0.9073
Gradient Boosting	0.9202	0.9114	---	---	0.9056
Random Forest	0.9846	0.8984	---	---	0.8915
Decision Tree	0.9997	0.8069	5.88	7.60	0.8012

** Best model*

BEST GENERALISATION

Linear Regression

Test R2 0.9156 — outperforms complex models

MOST OVERFIT

Decision Tree

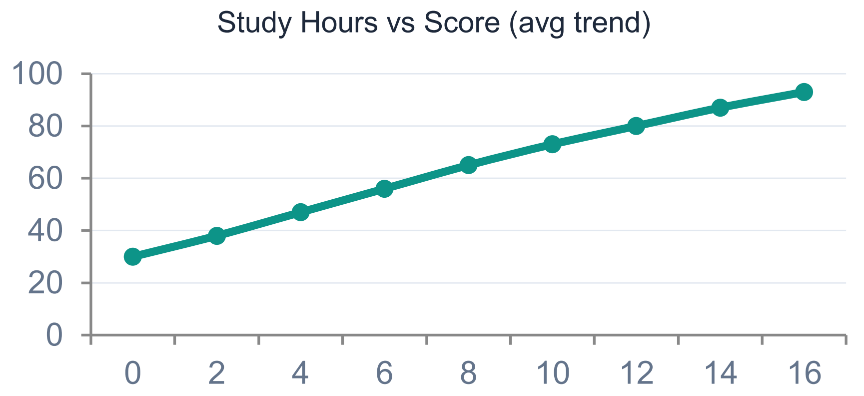
Train 0.9997 vs Test 0.8069 — clear overfit

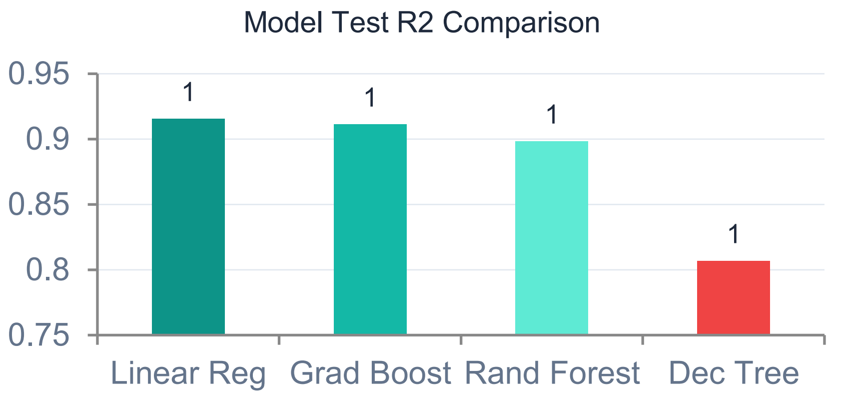
BEST CV STABILITY

Linear Regression

5-fold avg R2 = 0.9073 — most consistent

MODEL EVALUATION VISUALS





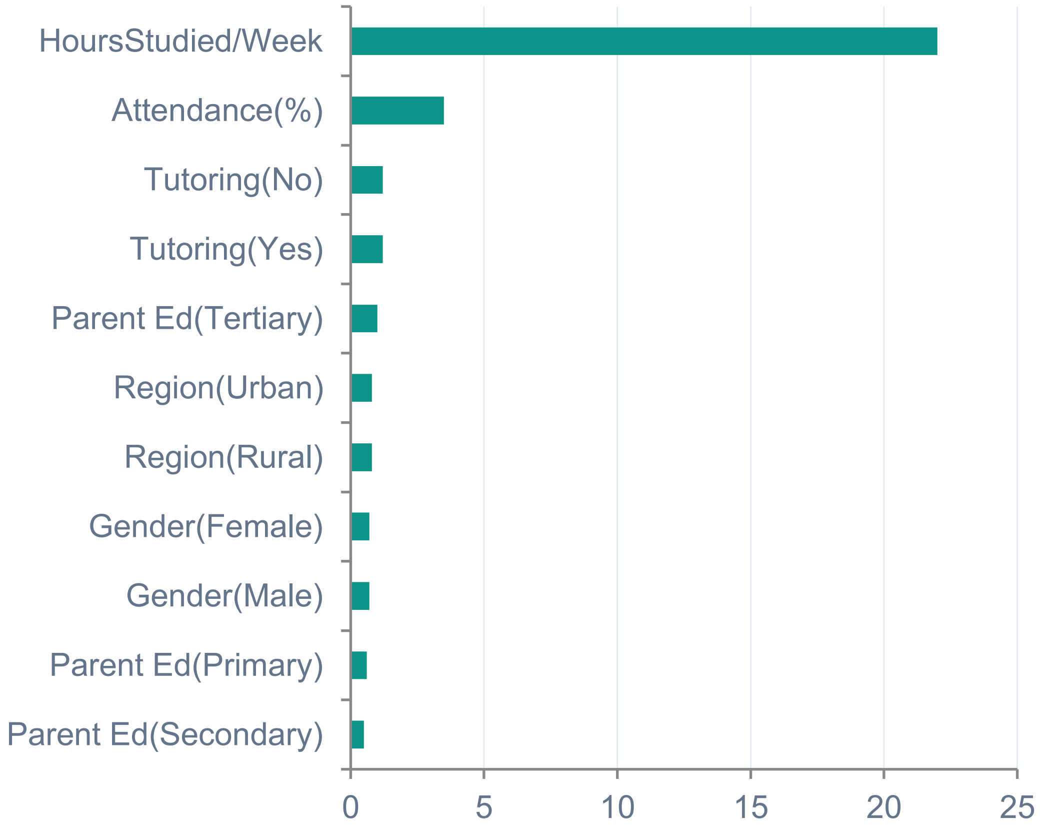
LEARNING CURVE INSIGHT

Training score (0.922) converges toward validation (~0.908) as data increases. The narrowing gap confirms no significant overfitting. Model generalises well to new students.

PREDICTION ERROR DISTRIBUTION

Errors are approximately normal and centred at zero, confirming unbiased predictions. The majority of predictions fall within +/-5 marks. Outlier tails extend to +/-15.

FEATURE IMPORTANCE - SHAP ANALYSIS



DOMINANT **HoursStudied/Week**

Each extra study hour strongly increases predicted score. SHAP values span -35 to +25.

MODERATE **Attendance(%)**

Secondary driver. Higher attendance mildly improves score — but far less than study time.

SMALL **Tutoring**

Small positive signal. Students with tutoring score slightly higher on average.

MINIMAL **Gender / Region / Parent Ed**

Near-zero SHAP values. Demographic features have negligible predictive power here.

CONCLUSIONS & FUTURE WORK

KEY CONCLUSIONS

> Study hours/week is the dominant predictor — r = 0.82 with exam score

> Linear Regression outperforms all complex models (Test R2 = 0.9156)

> Predictions are unbiased — error distribution is normal and centred at zero

> Demographic features (gender, region) have negligible predictive power

FUTURE WORK

-> Collect deeper features: sleep, motivation, socioeconomic index

-> Deploy as a web app (Flask / Streamlit) for real-time predictions

-> Explore neural networks on a larger dataset (10k+ records)

-> Build early-warning alerts for at-risk students

91.6%	3.99	5.03	0.907
R2 Score	MAE	RMSE	5-fold CV R2